

SHANJAI PUSHPARAJ

Electrical & Electronic Engineering Graduate | Embedded Systems | IoT | Robotics

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering undergraduate (SLIIT, expected Oct 2026) with hands-on embedded systems experience spanning **ESP32/STM32 development, PCB design, TinyML deployment, and wireless sensing**. Completed a 12-week industrial R&D internship prototyping and validating a LoRa-based current monitoring system and unit-level QC testing. Seeking entry-level roles in embedded systems, IoT, or hardware R&D.

EDUCATION

BSc (Hons) in Electrical and Electronic Engineering | Sri Lanka Institute of Information Technology, Malabe. *Expected Oct 2026*
Relevant coursework: Embedded Systems, Robotics, Power Electronics, Electrical Machines, Control Systems, Digital Electronics, Microcontrollers, Communication Systems, Signal Processing

Cambridge Advanced Level — Physical Sciences | Lyceum International School, Nugegoda *June 2022*

Cambridge Ordinary Level | Lyceum International School, Nugegoda *June 2020*

EXPERIENCE

Engineering Intern — Research & Development | Volt Charge Sri Lanka, JAT Holdings PLC *Jun – Sep 2025*

- Prototyped a **LoRa-based current monitoring system** (ESP32, RA-01 LoRa, CT clamps) with CRC16 error detection, achieving reliable wireless transmission over **~100 m** using optimized 433 MHz antennas.
- Calibrated RMS current readings against an **ADL400-D reference energy meter** for load monitoring and dynamic load-balancing support.
- Built and validated an **8-channel electrical QC testing table**, increasing simultaneous unit-test throughput.
- Ran unit-level QC: firmware upload, device ID configuration, relay/LED/RCM testing, and endurance testing across production units.
- Gained hands-on exposure to soldering, SMD rework, PCB repair, oscilloscope diagnostics, and electrical safety validation.

Engineering Intern | Melwire Rolling Pvt Ltd, Ekala *Jun – Sep 2024*

- Supported maintenance and troubleshooting of electrical systems in a steel manufacturing plant.
- Assisted a **33 kV solar power installation** project through data collection and on-site implementation support.
- Gained exposure to PLC-based automation and ladder programming in an industrial setting.

Teaching Assistant — Mathematics & Physics | Anapheema Institution, Wattala *Jan 2024 – Present*

- Tutor Cambridge O-Level students, translating technical concepts into clear, exam-focused instruction.

SELECTED PROJECTS

EEG-Based Wearable System for Sleep Quality Assessment | STM32H743VIT6, ESP32, ADS1220, TinyML, BLE, Flutter

- Deployed an INT8 TinyML sleep-staging model achieving **83.79% test accuracy in just 22 KB flash / 1.67 KB RAM** using ST Edge AI Studio.
- Designed the custom **4-layer STM32H743VIT6 PCB** — schematic, routing, grounding, and power distribution — in EasyEDA.
- Built the full embedded pipeline: EEG acquisition → DSP filtering → feature extraction → UART/SPI comms → TinyML inference → BLE transmission to a Flutter app.

LoRa-Based Current Monitoring System | ESP32, RA-01 LoRa, CT Clamp, PlatformIO, KiCad

- Designed CT signal conditioning (burden resistor selection, 1.65 V ADC biasing, filtering) and a compact LoRa packet format with CRC16 validation.

Smart Server Rack Monitoring System | ESP32, LabVIEW, Sensor Fusion

- Built real-time multi-point temperature/humidity/voltage monitoring with LabVIEW visualization, rack health index, and threshold-based alarms.

Paddy Bug Detection & Repellent System | ESP32, INMP441, LoRa, FFT

- Designed acoustic pest detection using IIR filtering and FFT-based signal analysis with LoRa pole-to-pole field communication.

Line Follower Robot & Wi-Fi RC Car | ESP32, Fuzzy Logic, TB6612FNG, L298N

- Built a fuzzy-logic-controlled line follower and a Wi-Fi web-controlled RC car for autonomous/remote motor control.

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, Java, Arduino, Ladder Logic
Embedded & IoT: ESP32, STM32, PlatformIO, STM32CubeIDE, BLE, LoRa, UART/SPI/I2C, ADC signal measurement
PCB & Hardware: EasyEDA, KiCad, 4-layer PCB design/routing, soldering, SMD rework, oscilloscope diagnostics
Simulation & Tools: LabVIEW, MATLAB, NI Multisim, Quartus

LEADERSHIP & VOLUNTEERING

Vice President, Rotaract Club of Colombo Port City — RI Year 2026/27
Vice President, Rotaract Club of SLIIT — RI Year 2025/26
Director of Public Relations, Rotaract Club of SLIIT — RI Year 2024/25

ACHIEVEMENTS

- District Commissioner's Award in Scouting
- Runner-up, Inter-International U-19 Cricket Tournament
- All Island Prize Winner, Spoken and Written English (IWMS)
- 2nd Runner-up, Electrical Category — Lyceum Annual Life Skills Day

LANGUAGES

English (Fluent) | Tamil (Fluent) | Sinhala (Professional Working Proficiency)

REFERENCES

Mr. Sanka Indushan — Senior Technology Entrepreneur, MAS Capital (Pvt) Ltd | +94 766 780 346 | sankai@masholdings.com
Dr. Nushara Wedasingha — Lecturer, Dept. of Electrical & Electronic Engineering, SLIIT | 076 449 2051 | nushara.w@sliit.lk